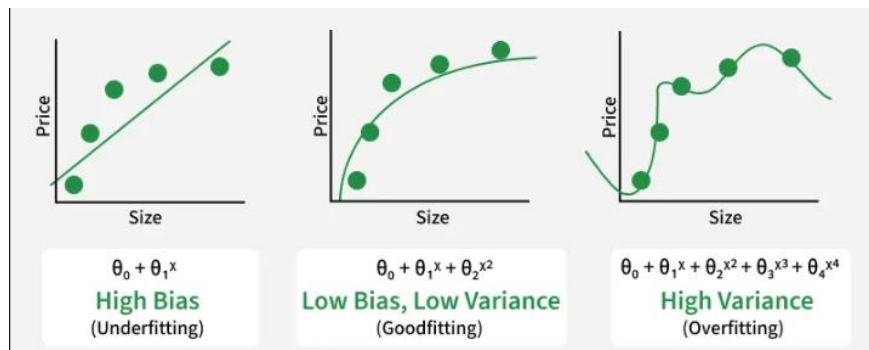


Overfitting vs Underfitting:

- Underfitting is when the model is way too simple to cover all the patterns in real datasets. So, it performs poorly on both the training and new unseen datasets.
- Overfitting is when the model is overly trained to the point that it memorizes the patterns, noise values and values out of the range, which results in a perfect performance on training data but poor performance on new unseen data.
- A good model is one that's not too simple yet not too complex. It should cover the important patterns and not memorize the noises and patterns.



- Causes of overfitting and underfitting:
 - Insufficient training data: the model doesn't find enough samples to learn properly from
 - Noisy data: datasets with many outliers, errors and irrelative values
 - Poor data quality: datasets with many duplicates and missing values
 - Unrepresentative data: the training data doesn't reflect the real world, so it's data is biased